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### Systems, devices, and related methods for fastening tissue

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#### Abstract

A fastening device may comprise a flexible longitudinal body including a central longitudinal axis and a distal end; an anvil extending distally from the distal end of longitudinal body; and a distal body extending distally from the distal end of the longitudinal body. The distal body may be configured to removably receive a cartridge having a plurality of fasteners. At least one of the anvil and the distal body may be flexible and may have a first configuration with a distal portion of the at least one of the anvil and the distal body extending radially outward from the central longitudinal axis and a second configuration with the distal portion closer to the central longitudinal axis than in the first configuration.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application is a National Stage entry under 35 U.S.C. § 371 of International Application No. PCT/US2020/056472, filed Oct. 20, 2020, which claims the benefit of priority from U.S. Provisional Application No. 62/924,557, filed Oct. 22, 2019, each of which is incorporated by reference herein in its entirety.

## TECHNICAL FIELD

(1) Various aspects of the present disclosure relate generally to tissue fastening, including visualizing, retracting, and coupling tissue. More specifically, at least certain embodiments of the present disclosure relate to systems, devices, and related methods for stapling tissue, among other aspects.

## BACKGROUND

(2) Technological developments have given users of medical systems, devices, and methods, the ability to conduct increasingly complex procedures on subjects. The coupling of tissue in, for example, a subject's gastrointestinal tract, is a type of procedure in which difficulties may arise. One such difficulty involves replacing an empty staple cartridge with a new cartridge full of staples during a procedure. When a user deploys all of the fasteners stored in a cartridge of a surgical fastener, the user may need to remove the device from the patient's body, replace the cartridge with a cartridge full of fasteners, and re-insert the surgical fastener into the patient's body to continue fastening tissue. By removing and re-inserting the stapling device into the body of the patient, the user may cause damage to the body of the patient caused by the additional movement of the device through the patient's body required to replace the stapling cartridge. Also, the time of the procedure may be increased due to the processes of removing and re-inserting the stapling device in order to replace the cartridge of staples. There is a need for stapler instruments that address this difficulty.

## SUMMARY

(3) Aspects of the disclosure relate to, among other things, systems, devices, and methods for fastening tissue. Each of the aspects disclosed herein may include one or more of the features described in connection with any of the other disclosed aspects.

(4) A fastening device may comprise a flexible longitudinal body including a central longitudinal axis and a distal end; an anvil extending distally from the distal end of longitudinal body; and a distal body extending distally from the distal end of the longitudinal body. The distal body may be configured to removably receive a cartridge having a plurality of fasteners. At least one of the anvil and the distal body may be flexible and may have a first configuration with a distal portion of the at least one of the anvil and the distal body extending radially outward from the central longitudinal axis and a second configuration with the distal portion closer to the central longitudinal axis than in the first configuration.

(5) Any of the devices disclosed herein may have any of the following features. The fastening device may include a protrusion extending from a surface of anvil, and the protrusion may be configured to engage a distal body. The at least one of the anvil and the distal body may be the anvil; and the distal body may be flexible and have a first configuration with a distal portion of the distal body extending radially outward from the central longitudinal axis and a second configuration with the distal portion of the distal body closer to the central longitudinal axis than in the first configuration. Each of the distal body and the anvil may be curved in the respective first configuration and substantially straight in the second configuration. The cartridge may be removably coupled to the distal body. The distal body may include a groove configured to slidably receive a flange of the cartridge. The cartridge may include an actuation wire extending proximally from a proximal end of the cartridge through a channel of the longitudinal body, and the actuation wire may be configured to move proximally relative to the cartridge and the anvil to deploy fasteners from the cartridge. A distal portion of the cartridge may include a locking portion configured to be received by a locking gap of the distal body. The distal body may include a first distal body and a second distal body spaced from the first distal body. The cartridge may include a

distal protrusion extending distally from a distal surface of the cartridge, and the distal protrusion may be configured to be received between distal ends of the first distal body and the second distal body. The distal body may include a first distal body and a second distal body separated by a gap configured to receive the cartridge. The device may further include a flexible catheter configured to translate relative to the longitudinal body, and the flexible catheter may move one or more of the anvil and the distal body towards each other when the flexible catheter is moved distally relative to the longitudinal body. The device may further include an I-beam including a top portion, a bottom portion, a distal-facing sharp edge and at least one actuation ramp, and the cartridge. The cartridge may be configured to receive the at least one actuation ramp. The I-beam may move proximally and distally relative to the anvil and the distal body. The distal-facing sharp edge may be proximal to a distal end of the I-beam and may extend between the top portion and the bottom portion of the I-beam. The device may further include a cartridge configured to hold a plurality of fasteners and including a proximal elongate extending proximally from a proximal end of the cartridge. The proximal elongate may include a lumen and at least one actuation wire positioned within the lumen. The actuation wire may be configured to deploy fasteners from the cartridge and/or actuate a knife within the cartridge. Each of the distal body and the anvil may include a radially-inward facing surface having at least one of a roughened surface, serrations, and teeth.

(6) In another example, a tissue fastening device may comprise a flexible tube and a fastener device positioned within a lumen of the flexible tube. The fastener device may comprise a flexible anvil that has a first configuration with a distal portion of the anvil extending radially outward from the central longitudinal axis of the fastener device and a second configuration with the distal portion of the anvil closer to the central longitudinal axis of the fastener device than in the first configuration. The fastener device may also comprise a flexible distal body configured to removably receive a cartridge of fasteners. The flexible distal body may have a first configuration with a distal portion of the flexible distal body extending radially outward from the central longitudinal axis of the fastener device and a second configuration with the distal portion closer to the central longitudinal axis of the fastener device than in the first configuration. The tube may be configured to translate distally relative to the fastener device to transition each of the anvil and the flexible distal body from the respective first configuration to the respective second configuration.

(7) Any of the devices disclosed herein may have any of the following features. The anvil may include a protrusion extending from a surface of the anvil. The protrusion may be configured to engage the flexible distal body. The cartridge may be removably coupled to the flexible distal body.

(8) A medical method may comprise inserting a flexible fastening device into a flexible tube. The fastening device may include an anvil, at least one distal body, and a cartridge removably coupled to the distal body and holding a plurality of fasteners. The method may also comprise inserting the tube into a natural orifice of a body. The method may further comprise moving the tube distally relative to the fastening device such that the tube moves at least one of the anvil and the distal body from a first configuration with a distal portion of the at least one of the anvil and the distal body extending radially outward from a central longitudinal axis of the fastening device to a second configuration with the distal portion closer to the central longitudinal axis of the fastening device than in the first configuration. The method may also comprise deploying at least one fastener from the cartridge.

(9) Any of the methods disclosed herein may include any of the following steps or features. The cartridge may be a first cartridge, and the method may further comprise uncoupling the first cartridge from the distal body, while the fastening device is in the body; removing the first cartridge from the body by moving the first cartridge proximally; and coupling a second cartridge to the distal body; wherein the second cartridge holds a plurality of fasteners.

(10) It may be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention, as claimed.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate exemplary aspects of the present disclosure and together with the description, serve to explain the principles of the disclosure.
- (2) FIG. 1 is a perspective view of an exemplary surgical stapler system, according to aspects of this disclosure.
- (3) FIGS. 2A and 2B are perspective and side views, respectively, of an exemplary medical device according to aspects of this disclosure.
- (4) FIG. 3 is a perspective view of a portion of an exemplary medical device, according to aspects of this disclosure.
- (5) FIG. 4 is a perspective view of a portion of an exemplary medical device, according to aspects of this disclosure.
- (6) FIG. 5 is a perspective view of a portion of an exemplary medical device with part of the medical device shown in cross-section, according to aspects of the present disclosure.
- (7) FIGS. 6A and 6B are side views of an exemplary surgical stapler system, according to aspects of this disclosure.
- (8) FIG. 7 is a side view of a portion of an exemplary surgical stapler system, according to aspects of the present disclosure.
- (9) FIG. 8 is a side view of exemplary surgical stapler system, according to aspects of this disclosure.
- (10) FIG. 9 is a perspective view of an exemplary surgical stapler system, according to aspects of this disclosure.
- (11) FIG. 10 is a perspective view of an exemplary component of a surgical stapler system, according to aspects of this disclosure.
- (12) FIG. 11 is a perspective view of an exemplary component of a surgical stapler system, according to aspects of this disclosure.

### DETAILED DESCRIPTION

(13) The present disclosure is drawn to systems, devices, and methods for coupling, cutting, and resecting tissue, among other aspects. Reference will now be made in detail to aspects of the present disclosure, examples of which are illustrated in the accompanying drawings. Wherever possible, the same or similar reference numbers will be used through the drawings to refer to the same or like parts. The term “distal” refers to a portion farthest away from a user when introducing a device into a patient. By contrast, the term “proximal” refers to a portion closest to the user when placing the device into the patient. The term “coupling tissue together” may refer, for example, to stapling, fixing, attaching, fastening, or otherwise joining two portions of tissue together. The term “fastener” may include staples, clips, elastic bands, suture, or any other fastener known in the art. As used herein, the terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not necessarily include only those elements, but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. The term “exemplary” is used in the sense of “example,” rather than “ideal.”

(14) Embodiments of the present disclosure may be used to visualize, cut, resect, and/or couple together target tissue in an endo-luminal space, or facilitate the process thereof. In particular, some embodiments combine a tissue resecting device with a tissue stapling device. The tissue stapling apparatus may include a resection or cutting mechanism (e.g., an integrated knife or sharp edge) and a stapling mechanism (stapler). The stapling apparatus may be delivered to target tissue through an endoscope working channel to the target tissue site. In some examples, the stapling

apparatus may be back-fed through an endoscope, gastroscope, colonoscope, flexible catheter, or other medical device working channel prior to inserting the device into the body of the patient. The overall system may include a retraction mechanism, such as a tissue clip, to pull tissue towards the tissue stapling device. All or parts of the tissue stapling device and the retraction mechanism could be metallic, composite, plastic, or include a shape memory metal (such as nitinol), a shape memory polymer, a polymer, or any combination of materials.

(15) FIG. 1 shows a distal end of an exemplary surgical system **100** in accordance with an embodiment of this disclosure. System **100** may include a surgical stapling apparatus **101** configured to engage body tissue, apply a plurality of surgical fasteners thereto, and form an incision in the fastened body tissue. Surgical stapling apparatus **101** may be used during minimally invasive surgical procedures, such as laparoscopic or endoscopic procedures, or any other suitable medical procedure. Apparatus **101** may be used to apply surgical clips or other fasteners, but will be primarily discussed in the context of applying staples from a staple cartridge. Apparatus **101** is shown positioned within a flexible catheter **103** including a working channel **105** in which an elongate body **102** of apparatus **101** is positioned.

(16) As illustrated in FIG. 1, elongate body **102** of apparatus **101** may be at least partially cylindrical, flexible, and configured to move through a working channel **105** of a medical device, such as catheter **103**. Elongate body **102** may extend longitudinally from a proximal end (not shown) to a distal end (shown in FIG. 1). An anvil **104** and two distal bodies **106**, **107** may extend distally from a distal portion of elongate body **102**. Anvil **104** and distal bodies **106**, **107** may extend from opposing portions of elongate body **102** and, in one configuration of bodies **106**, **107**, may curve or be biased radially outward in opposite directions from the longitudinal axis of elongate body **102** in a first, e.g., unconstrained or open, configuration. When catheter **103** slides distally over body **102**, distal bodies **106**, **107** and anvil **104** may move radially inward towards the central longitudinal axis of apparatus **101**. In some examples, when catheter **103** slides distally over body **102**, distal bodies **106**, **107** may be moved towards anvil **104** and transition from the first configuration to a second, e.g., constrained or closed, configuration. In some examples, a user may position target tissue between anvil **104** and distal bodies **106**, **107** so that when catheter **103** slides distally over body **102**, anvil **104** and distal bodies **106**, **107** clamp down onto the target tissue. When positioned in the second, e.g. constrained or closed configuration, teeth **122** of apparatus **101** may engage target tissue. A channel **140** (shown in FIG. 2) may extend longitudinally within a radially-outer surface of elongate body **102**. At least proximal portions of radially-inward facing surfaces, relative to the longitudinal axis of elongate body **102**, of each of anvil **104** and distal bodies **106**, **107** may face each other. Anvil **104** and distal bodies **106**, **107** may be flexible and at least distal portions thereof may be biased radially-outward from the longitudinal axis of elongate body **102**. Such portions of anvil **104** and distal bodies **106**, **107** may be configured to flex inward towards the central longitudinal axis of elongate body **102** such that anvil **104** and distal bodies **106**, **107** are closer to or the same distance from the central longitudinal axis of elongate body as the radially-inner surface of elongate body **102**. Anvil **104** and distal bodies **106**, **107** may be spring steel, shape memory alloy, or any other material capable of being springing back or otherwise returning to its original shape after bending. In some examples, elongate body **102** may be coupled to an actuation wire, cable, or tube (not shown) at a proximal portion of elongate body **102**, and movement of the actuation wire, cable, or tube proximally or distally may move elongate body **102** proximally or distally.

(17) Anvil **104** may be configured to align with distal bodies **106**, **107** when distal bodies **106**, **107** are moved towards anvil **104**. Anvil **104** may include a textured portion, such as teeth or serrations **122**, at a distal portion of anvil **104**. In some examples, anvil **104** may include recesses **120** and a protruding portion **118** at a distal end of anvil **104**. Teeth or serrations **122** of anvil **104** may align with a textured portion, such as teeth or serrations **124**, of distal portions **106**, **107** when anvil **104** and distal portions **106**, **107** are positioned substantially parallel to a longitudinal axis of elongate

body **102**. Anvil **104** has substantially flat upper and lower surfaces, a rounded or curved distal tip portion, and a proximal portion extending from a distal edge of elongate body **102**. Recesses **120** of anvil **104** may be configured to received fasteners, such as staples. Recesses **120** may be configured to align with openings **114** of cartridge **110**. Protruding portion **118** may extend radially-inward relative to the central longitudinal axis of elongate body **102** from the rounded distal tip portion of anvil **104**. Protruding portion **118** may be hook-shaped and may be configured to engage cartridge **110**. As shown in FIG. 4, protruding portion **118** may include an edge portion **140** terminating at a sharp point at an end of protruding portion **118**. Edge portion **140** may be configured to pierce tissue so that protruding portion **118** may extend through tissue to engage or couple to at least one of distal bodies **106**, **107** and cartridge **110** to hold apparatus **101** in a locked or clamped position, which may facilitate deployment of fasteners. A radially-outer surface **144** of anvil **104** may be configured and positioned to engage a surface of catheter **103**, such as a surface defining lumen **105**.

(18) Distal bodies **106**, **107** may have longitudinal axes parallel to each other and may curve towards each other at their distal ends **108**, **109**. In some examples, distal ends **108**, **109** may be spaced from each other to provide a gap to receive a distal protrusion **128** of a stapler cartridge **110** (shown in FIG. 3). Each distal body **106**, **107** may include textured protrusions, such as a serrated or teeth portions **124** positioned at a distal portion of distal bodies **106**, **107**. Serrated or teeth portions **124** may be configured to grip tissue and may be positioned at or proximate to an outer edge **125** (shown in FIG. 2) of distal bodies **106**, **107**. Each distal body **106**, **107** may be spaced from the radially outermost surfaces **146** of elongate body **102**. Radially-outer surfaces of distal bodies **106**, **107** may be configured and positioned to engage a surface of a working channel of a medical device, such as a surface defining lumen **105** in catheter **103**. Anvil **104** may extend from a distal edge of elongate body **102** and a proximal space **133** between distal bodies **106**, **107** and anvil **104** may be less than the diameter or cross-sectional dimension of elongate body **102**.

(19) Any of the embodiments disclosed herein may have only one of anvil **104** or distal bodies **106**, **107** that curves radially outward, is flexible, and transitions between a configuration spaced from the longitudinal axis of elongate body **102** (to accept tissue) to a stapling configuration that may be substantially parallel to the axis. The other (anvil **104** or distal bodies **106**, **107**) may extend substantially parallel to the longitudinal axis of elongate body **102**, may be straight and/or substantially inflexible, and may not transition between spaced and stapling configurations.

(20) Distal bodies **106**, **107** may be configured to hold a fastener cartridge **110**, as shown in FIG. 1. FIG. 5 shows a perspective view of distal bodies **106**, **107** with a stapler cartridge **110** positioned within grooves **130**, **132** of distal bodies **106**, **107**, with portions of distal bodies **106**, **107** and cartridge **110** shown in a cross-section. Each groove **130**, **132** of distal bodies **106**, **107** may extend from distal ends **108**, **109** to a proximal portion of elongate body **102**. Groove **130** of distal body **106** may be positioned on a first surface **157** opposing a second surface **158** of distal body **107** in which groove **132** is positioned. In some examples, grooves **130**, **132** may extend to the proximalmost end of elongate body **102**. Grooves **130**, **132** may be configured to slidably receive protrusions or tangs **131**, **133** respectively of stapler cartridge **110**.

(21) In some examples, distal bodies **106**, **107** may include a locking gap portion **190** between a distal portion of each distal body **106**, **107**. Opposing surfaces **198**, **199** surrounding locking gap portion **190** of distal bodies **106**, **107** may be spaced farther apart than opposing surfaces, such as first surface **157** and second surface **158**, at a proximal portion of distal bodies **106**, **107**, and ledges **195**, **196** may be formed on distal bodies **106**, **107**. Ledges **195**, **196** may be configured to receive a locking portion **191** of cartridge **110**. In some examples, grooves **130**, **132** may end at ledges **195**, **196** (e.g. extending from a proximal end to ledges **195**, **196**). A further discussion of how locking gap portion **190** may prevent movement of cartridge **110** and may facilitate positioning of cartridge **110** is described below. In some examples, distal bodies **106**, **107** may not include locking gap portion **190**.

(22) Cartridge **110** may be configured to be positioned between distal bodies **106**, **107**. Cartridge **110** may include openings **114** configured to deploy fasteners and a longitudinal slot **116** configured to receive a cutting device, such as a knife. A distal portion **126** of cartridge **110** may be angled relative to the longitudinal axis of cartridge **110**. Cartridge **110** may also include tangs **131**, **133** extending transverse to a longitudinal axis of cartridge **110**. Tangs **131**, **133** may be sized to be received within grooves **130**, **132**. In some examples, tangs **131**, **133** may be sized to fit within grooves **130**, **132** and allow sliding movement of cartridge **110** in the proximal-distal directions and slight side-to-side and/or up-to-down movement in a direction transverse to the proximal-distal directions. In some embodiments, the tangs **131**, **133** and/or the grooves **130**, **132** may include a taper, so that the tangs **131**, **133** may form a secure fit in the grooves **130**, **132** when positioned in the distal bodies **106**, **107**. For example, pushing the cartridge **110** distally relative to the distal bodies **106**, **107**, may engage and secure the cartridge **110** in a desired position. When the cartridge **110** should be removed and/or replaced, the cartridge can be manipulated by pulling proximally relative to the distal bodies **106**, **107** to disengage the tapered portions of the tangs **131**, **133** and/or the grooves **130**, **132**. In addition, cartridge **110** may include one or more cavities **150**, **151** (shown in FIG. 5). Cavities **150**, **151** may be configured to receive one or more actuators, one or more fasteners, one or more pistons or spacers, and/or one or more knives. A proximal elongate **112** may extend from a proximal portion of cartridge **110**. Proximal elongate **112** may be cylindrical and may include a lumen (not shown) for receiving one or more actuation wires for deploying fasteners and/or for moving one or more cutting devices (such as a knife). Proximal elongate **112** may be flexible and may be configured to move through elongate body **102**. Proximal elongate **112** may be sufficiently rigid to translate cartridge **110** distally when a user moves a proximal portion of proximal elongate **112** distally. In some examples, proximal end of proximal elongate **112** may include a spool style handle, a wheel shaped knob, and/or any other suitable activation mechanism known in the art, that deploys fasteners by either pulling or rotating. Cartridge **110** may include a distal protrusion **128**, and distal protrusion **128** may be configured to be positioned within gap **137** between distal portions **108**, **109** of distal bodies **106**, **107**.

(23) FIGS. 6A and 6B are side views of an exemplary surgical stapling apparatus **601** substantially similar to apparatus **101**. Any of the features discussed with regard to apparatus **101** may be included in apparatus **601**. Apparatus **601** is shown positioned within flexible catheter **603** (shown in cross-section). Apparatus **601** includes distal body **606**, anvil **604** including protruding portion **618**, staple cartridge **610**, elongate body **602**, and proximal elongate **612**. Apparatus **601** may also include a second distal body, like distal body **107**, but not shown in the views of FIGS. 6A-6B. In FIG. 6A, apparatus **601** is positioned with anvil **604** and distal body **606** separated in an expanded, open configuration. In some examples, apparatus **601** may be back-fed into flexible catheter **603**, which may include moving the proximal end (not shown) of apparatus **601** through distal opening **601** of flexible catheter **603** in order to position anvil **604** and distal body **606** proximate to distal opening **601**. A user may first position tissue within space **698** (the active region of apparatus **601**) between anvil **604** and distal body **606**. The user may then move flexible catheter **603** distally relative to apparatus **601** and cause curved outer surface portions of anvil **604** and distal body **606** to contact inner surfaces of flexible catheter **603**. By moving flexible catheter **603** distally relative to apparatus **601**, anvil **604** and distal body **606** may be moved towards each other to clamp tissue between anvil **604** and distal body **606**. In other embodiments, as discussed above, one of anvil **604** or distal body **606** may be substantially straight, may be substantially inflexible, may extend substantially parallel to the longitudinal axis of elongate body **602**, and may remain in the same position as flexible catheter **603** is moved distally, thus causing only the other of anvil **604** or distal body **606** to move radially inward as flexible catheter **603** is moved distally.

(24) In some examples, protruding portion **618** of anvil **604** may assist in holding anvil **604** and distal body **606** in a closed configuration (clamped together) to allow a user to deploy staples or other fasteners. For example, protruding portion **618** may include a flange **619**, and flange **619** may



engage cartridge **610** and/or an outer surface of distal body **606**, to hold distal body **606** (shown in FIG. **6B**) in a clamped position and prevent distal body **606** from moving radially-outward relative to the longitudinal axis of body **602**. By holding anvil **604** and distal body **606** clamped together, protruding portion **618** may allow the user to stop moving flexible catheter **603** distally, moving apparatus **601** proximally, and/or applying a force in the distal direction to catheter **603** (in order to hold anvil **604** and distal body **606** together) since anvil **604** and distal body **606** will be held in a closed, clamped position. The user may then proceed to deploy staples from cartridge **610**. In some examples, the user may pull an actuator, which may be positioned within proximal elongate **612**, to deploy staples from cartridge **610** when anvil **604** and distal body **606** are clamped together (shown in FIG. **6B**). An example of such activation will be described herein with respect to FIG. **7**. To release protruding portion **618** from cartridge **610**, the user may move flexible catheter **603** distally to contact flange **619** and move flange **619** distally to release cartridge **610** from contacting flange **619**. Since distal body **606** may be biased towards an open position shown in FIG. **6A**, when flange **619** is moved distally out of contact with cartridge **610**, distal body **606** will move radially outward and away from anvil **604**. Alternatively, the user may pull the staple cartridge **610** proximally to move flange **619** out of contact with the staple cartridge **610**. In some examples, anvil **604** may not include a protruding portion **618**, and the user may hold anvil **604** and distal body **606** in a closed, clamped position by holding flexible catheter **603** in a position that forces anvil **604** and distal body **606** in a clamped position. While apparatus **601** is shown positioned within flexible catheter **603**, apparatus **601** may operate in the same manner as described above when positioned within a working channel of another type of medical device, such as a working channel of an endoscope, gastroscope, colonoscope, ureterscope, or the like.

(25) FIG. **7** shows an enlarged, cross-sectional view of an exemplary stapling apparatus **701** including anvil **704**, distal body **706**, cartridge **710** with fasteners **745** (staples), and an actuation sled **741** with actuation body **743**. Any of the components of apparatus **701** may include any of the features discussed herein regarding the components of other stapling apparatuses **101**, **601**. The components shown within cartridge **710** may be positioned with cavity **150**, cavity **151**, a single cavity within a cartridge, or a plurality of cavities within a cartridge. The apparatus **701** in FIG. **7** is shown deploying fasteners **745** onto tissue **749**. Actuation sled **741** engages pistons or spacers **751**, and the pistons or spacers **751** then engage the fasteners **745** to deploy the fasteners **745**. As shown in FIG. **7**, actuation sled **741** engages the pistons or spacers **751** as actuation sled **741** moves in the proximal direction **P**. Each piston or spacer **751** may be configured to translate within cartridge **710** and may be sized to engage only one fastener **345**. In other examples, each piston or spacer **751** may be sized to engage multiple fasteners **345**, or one or more pistons or spacers **751** may be sized to engage multiple fasteners **345** while one or more other pistons or spacers **751** may be sized to engage only one fastener **345**. Actuation sled **741** may be configured to move each piston or spacer **751** in a direction transverse to the longitudinal axis of distal body **706**.

(26) For example, actuation sled **741** and actuation body **743** (which may be a wire or cable, extending to a proximal handle) may be pulled proximally, which may move ramp **742** (including first portion **790** and second portion **791**) into contact with one or more pistons or spacers **751** and push one or more pistons or spacers **751** to deploy one or more fasteners **745** from cartridge **710**. First portion **790** of ramp **742** may have a steeper incline relative to second portion **791**. Each of the pistons or spacers **751** may have an upper surface that aligns and/or is flush with a portion of each respective fastener **745**. For example, each fastener **745** may be a staple and may include three substantially flat sections (e.g., a “U-shape”) with the middle substantially flat section aligning with the top surface of each spacer **751**. Each piston or spacer **751** may be sufficiently rigid to move uniformly upward when the ramp **742** of actuation sled engages a corner of the piston or spacer **751**. In some examples, each piston or spacer **751** may be coupled to cartridge **710** such that movement along the longitudinal axis of cartridge **710**, or in the proximal **P** or distal **D** directions shown in FIG. **7**, is prevented while allowing movement in a direction perpendicular to the

longitudinal axis of cartridge **710**. The pistons or spacers **751** in cartridge **710** may prevent fasteners **745**, such as staples, from partially deploying and may avoid improper stapling caused by a distal portion of a staple moving upward when a proximal portion of the staple does not move. By providing spacers **751** with flat top surfaces that align with fasteners **745**, the ramp **742** of actuation sled **741** can move proximally and push each fastener **745** uniformly upward by engaging each spacer **751**. When fasteners **745** are deployed from cartridge **710**, fasteners **745** may engage recesses **760** of anvil **704** (for example) to facilitate fastening to tissue **749**.

(27) After the user has deployed all of the fasteners **745** from cartridge **710**, the user may then remove cartridge **710** from distal body **706** and replace cartridge **710** with a new cartridge full of fasteners. For examples, referring to FIGS. **1** and **5**, the user may move cartridge **110** proximally by pulling proximal elongate **112** proximally, and cartridge **110** may slide proximally through elongate body **102** within grooves **130**, **132** and be removed by the user at a proximal portion of elongate body **102**. Once completely removed from apparatus **101**, the user may replace cartridge **110** with a new cartridge full of fasteners. The user may then position the new cartridge within grooves **130**, **132** and move the new cartridge distally to distal bodies **106**, **107**. Once the new cartridge is positioned at the distal end of distal bodies **106**, **107**, the user may continue fastening tissue using the new cartridge. By providing a way to remove and replace staple cartridges without having to remove the entire stapling apparatus **101** from a working channel, the user may avoid or minimize unnecessary damage to the patient caused by the removal and re-insertion of apparatus **101** out of and into the patient's body during a procedure. Also, replacing an empty staple cartridge with a new cartridge without having to remove the entire stapling apparatus **101** from the patient's body may reduce procedure time.

(28) In some examples, any of the cartridges **110**, **610**, **710** discussed herein may include a locking portion **191**. Locking portion **191** may be configured to be received by a locking gap portion **190** between distal bodies **106**, **107**. Locking portion **191** may be positioned between distal bodies **108**, **109** and tangs **131**, **133** may be positioned within grooves **130**, **132** of distal bodies **106**, **107**. An outer portion of locking portion **190** that is radially-outermost from the central longitudinal axis of cartridge **110** may be positioned adjacent to and slidably engaged with distal bodies **106**, **107** as cartridge is moved from a proximal portion of distal bodies **106**, **107** to locking gap portion **190**. For example, cartridge **110** may be moved distally or proximally with tangs **131**, **133** positioned within grooves **130**, **132** while locking portion **191** is positioned adjacent to distal bodies **106**, **107**. When cartridge **110** is positioned such that locking portion **191** is distal to ledges **195**, **196** of distal bodies **106**, **107**, locking portion **191** may be moved upward (or away from anvil **104**) to a position between distal bodies **106**, **107** within locking gap portion **190**. Once locking portion **191** is positioned within locking gap portion **190**, ledges **195**, **196** may prevent cartridge **110** from moving proximally, and distal portions **108**, **109** of distal bodies **106**, **107** may prevent cartridge **110** from moving distally. In some examples, distal protrusion **128** may provide a visual notification to the user when locking portion **191** is properly positioned within locking gap portion **190** because the user may identify when distal protrusion **128** is positioned within gap **137**, for example via an image sensor such as image sensor **864** described herein. When the user could like to remove cartridge **110** from locking gap portion **191**, the user may move cartridge **110** in a direction transverse to the longitudinal axis of distal bodies **106**, **107** until locking portion **191** is not in contact with ledges **195**, **196**, and then move cartridge **110** proximally.

(29) In some examples, cartridge **110** may be moved distally or proximally with tangs **131**, **133** of locking portion **191** positioned within grooves **130**, **132** such that distal bodies **106**, **107** are moved radially outward due to locking portion **191** being positioned between distal bodies **106**, **107**. Grooves **130**, **132** may extend to the distal ends of distal bodies **106**, **107**. When cartridge **110** is positioned such that locking portion **191** is distal to ledges **195**, **196** of distal bodies **106**, **107**, distal bodies **106**, **107** may move radially inward towards locking portion **191** to a position distal bodies **106**, **107** abutting locking portion **191** within locking gap portion **190**. Once locking portion **191** is

positioned within locking gap portion **190**, ledges **195**, **196** may prevent cartridge **110** from moving proximally, and distal portions **108**, **109** of distal bodies **106**, **107** may prevent cartridge **110** from moving distally.

(30) FIG. **8** shows an embodiment of a cartridge **810** and a proximal elongate member **861** extending out of a working channel **816** of distal tip **800** of a medical device, such as an endoscope or any other device mentioned herein. Distal tip **800** may include one or more image sensors **864** and one or more illuminators **865**. Cartridge **810** may be configured to operate with embodiments of stapler apparatuses similar to apparatus **101** or **601** that do not require the stapling actuator, such as actuation sled **741**, to be part of the cartridge **810**, such as stapling apparatus **901** discussed herein below. Cartridge **810** may include a coupling protrusion **867** configured to mate with coupler **863**. Surfaces of cartridge **810** may include cuts, slots, ridges, or other features that add flexibility to cartridge **810**, permitting cartridge **810** to more easily extend through working channel **816**. Similarly, proximal elongate member **861** may be flexible yet sufficiently rigid to move cartridge **810** through working channel **816**. Cartridge **810** may be coupled to proximal elongate member **861** via coupler **863**. In some examples, elongate member **861** may be configured to be back-fed through an endoscope working channel or other medical device prior to coupling cartridge **810** to elongate member **861**.

(31) Coupler **863** is shown in cross-section in FIG. **8**. In some examples, coupler **863** may mate with cartridge **810** and may be releasable via an actuator. In some examples, coupler **863** may be releasable without an actuator. The proximal end of elongate member **861** may have at least one user interface to release coupler **863** from cartridge **810**. Coupler **863** may be cylindrical, may cover portions of each of proximal elongate member **861** and cartridge **810**, and may include a lumen **875** extending through the central longitudinal axis of coupler **863**. In some examples, coupler **863** may include a radially-outer surface **890** that extends towards the central longitudinal axis of coupler **863** as the radially-outer surface moves from a longitudinal midpoint towards proximal end **891** and distal end **892** of coupler **863**. In some examples, a chamfered tip **868** of elongate member **861** may be configured to be received by coupler **863** to couple cartridge **810** to elongate member **861**. Chamfered tip **868** may be rigid and may be positioned within lumen **875** of coupler **863** (shown in FIG. **8**). Coupler **863** may be flush and/or in contact with the exterior surface of elongate member **861** at a portion proximal to the chamfered tip **868**. Chamfered tip **868** may facilitate breaking coupler **863** to release cartridge **810** from elongate member **861**. For example, a user may pull elongate member **861** proximally to pull coupler **863** proximally such that an exterior surface of coupler **863** contacts distal tip **800**, for example coupler **863** may contact a distal front face of distal tip **800** proximate to an edge portion of working channel **816**. When coupler **863** comes into contact with distal tip **800**, chamfered tip **868** may be pulled proximally such that chamfered tip **868** applies pressure against an interior surface of coupler **863**, which may compress coupler **863** via chamfered tip **868** moving proximally and coupler **863** abutting distal tip **800**. Compressing coupler **863** between distal tip **800** and chamfered tip **868** may cause coupler **863** to break and release cartridge **810** from coupler **863**. In some examples, coupler **863** may break into several pieces and be released into a body of a patient during a procedure. In some examples, coupler **863** may be cylindrical and may include an interior portion configured to receive proximal elongate member **861** and cartridge **810**. Coupler **863** may be overmolded and/or contain an overmolded feature in the shape of cartridge **810**. In some examples, an overmolded feature of coupler **863** may be overmolded over cartridge **810** such that once cartridge **810** is closed onto tissue to deploy staples from cartridge **810**, the tension applied to the entire assembly by the operator, due to stapling tissue using cartridge **810**, exceeds the material strength of coupler **863** and breaks coupler **863**, which may then release cartridge **810**. In some examples, coupler **863** may be biodegradable and/or digestible, and may be configured to break off from elongate member **861** and left within the body of a patient. In some examples, cartridge **810** may be manufactured with a portion of coupler **863** positioned within recess **871** of cartridge **810**. A user may use coupler **863**

and proximal elongate **861** to deliver cartridge **810** to a stapling apparatus, such as stapling apparatus **901**, couple cartridge **810** to stapling apparatus **901**, and release cartridge **810** from coupler **863**. In some examples, coupler **863** may be configured to break off from cartridge **810** after cartridge is fastened to or clamped down on tissue. In some examples, elongate member **861** and coupler **863** may be used to retrieve an empty staple cartridge from an apparatus, such as apparatus **901**. Cartridge **810** may include teeth on a surface of cartridge **810** configured to mate with one or more distal bodies, such as distal bodies **906**, **907**, to facilitate coupling cartridge **810** to the one or more distal bodies.

(32) In some examples, a user may deliver cartridge **810** to a distal portion of distal bodies **906**, **907** (shown in FIG. 9) using proximal elongate **861** and coupler **863**. A user may move proximal elongate **861** to coupler cartridge **810** to distal bodies **907**, **908**, which in some examples may be a snap-fit. Once cartridge **810** is fixedly coupled to distal bodies **907**, **908**, the user may pull proximal elongate **861** proximally. When pulling proximal elongate **861** proximally, chamfered tip **868** may apply pressure against the interior surface of coupler **863** because cartridge **810** is preventing coupler **863** from moving proximally with proximal elongate **861**. In some examples, the pressure created from the user pulling proximal elongate **861** proximally and chamfered tip **868** contacting the interior surface of coupler **863** may cause coupler **863** to break and release cartridge **810** from coupler **863**. In some examples, cartridge **810** may remain fixedly coupled to distal bodies **907**, **908** after coupler **863** releases from cartridge **810**.

(33) FIG. 9 shows another embodiment of a stapling apparatus **901** including an anvil **904** and distal bodies **906**, **907** positioned within a flexible catheter **903**. Apparatus **901** may have any of the features previously described in relation to the other apparatuses of this disclosure. Apparatus **901** may include an I-beam **964** partially surrounding at least proximal portions of anvil **904** and distal bodies **906**, **907**. Anvil **904** and distal bodies **906**, **907** may be curved and biased to an open position (shown in FIG. 9) in the same manner as distal bodies **106**, **107** and anvil **104**. Anvil **904** may include a first anvil body **960** and a second anvil body **961** that substantially align with distal bodies **906**, **907**, respectively. Proximal portions of each of first anvil body **960**, second anvil body **961**, and distal bodies **906**, **907** may be positioned between top portion **970** and bottom portion **971** of I-beam **964**. I-beam **964** may be positioned within a lumen of flexible catheter **903** or a working channel of an endoscope or other medical device. I-beam **964** may be coupled to a flexible wire (not shown) at a proximal portion of I-beam **964**. The flexible wire may extend proximally to a handle/actuator that may be activated by a user to move I-beam **964** proximally and distally relative to anvil **904** and distal bodies **906**, **907**. Cartridge **910** may be coupled to distal bodies **906**, **907** via a protrusion (not shown) fit between distal bodies **106**, **107** forming a snap-fit coupling, one or more hooks, or other coupling mechanisms including those described in other embodiments herein. In some examples, cartridge **910** may be coupled to distal bodies **906**, **907** via protrusions extending from cartridge **910** received within grooves of distal bodies **906**, **907**, in substantially the same manner that grooves **130**, **132** may slidably receive protrusions or tangs **131**, **133** of stapler cartridge **110** (as shown in FIG. 5).

(34) I-beam **964** may include a top portion **970**, a bottom portion **971**, an intermediate portion **965** connecting the top portion **970** to the bottom portion **971**, and two ramp actuators **968**, **969**. Top portion **970** and bottom portion **971** may be curved and may be configured to move within a working channel of a medical device or a lumen of flexible catheter **903**. In some examples, top portion **970** and bottom portion **971** may have substantially the same length and width relative to their longitudinal axes, may have substantially the same radius of curvature, and/or otherwise may be mirror images of one another. Intermediate portion **965** may be flat, with planar surfaces and a distal edge **966**. Distal edge **966** may be sharp and may be configured to cut tissue. Intermediate portion **965** may extend longitudinally substantially parallel with the longitudinal axes of top portion **964** and bottom portion **971**, and longitudinal ends of edge **966** may be coupled to radially-inner surfaces of top portion **964** and bottom portion **971**. In some examples, I-beam **964** may

include two ramp actuators **968, 969** configured to deploy fasteners from cartridge **910**. Ramp actuators **968, 969** may extend radially inward from a radially-inner surface of bottom portion **971** (shown in FIG. **10**) or top portion **970**. Each ramp actuator **968, 969** may include a radially-inward facing surface that extends proximally from a radially-inward facing surface of bottom portion **971**. Ramp actuators **968, 969** may be configured to align with channels of cartridge **910**. I-beam **964** may be configured to move via movement of the aforementioned wire coupled to a proximal portion of I-beam **964**.

(35) FIG. **11** shows an alternative embodiment of an I-beam **1164** with top portion **1170**, bottom portion **1171**, intermediate portion **1165**, edge **1166**, and a single ramp actuator **1168**. I-beam **1164** may have any of the characteristics previously described in relation to I-beam **964**. A single ramp actuator **1168** may extend distally from a radially-inward facing surface of bottom portion **1171** and may include a curved distal end.

(36) Cartridge **910** may include any of the features previously described in relation to cartridges **110, 610, 710, 810**. Cartridge **910** may include openings **914** configured to deploy fasteners and a longitudinal slot **916** configured to receive a cutting device, such as edge **966** of I-beam **964**. A distal portion **926** of cartridge **910** may be angled relative to the longitudinal axis of cartridge **910**. In addition, cartridge **910** may include one or more cavities configured to receive one or more actuation ramps **968, 971, 1168**. Cartridge **910** may be configured to align with distal bodies **906, 907**. In some examples, anvil **904** and/or distal bodies **906, 907** may extend to the proximal end of apparatus **901**. In some examples, the proximal ends of distal bodies **906, 907** may be coupled to control wires, and the control wires may be configured to be pulled proximally from the proximal end to move anvil **904** and distal bodies **906, 907** to a closed position.

(37) A user may use apparatus **901** to couple one or more fasteners to tissue by first positioning tissue between anvil **904** and distal bodies **906, 907**. In some examples, a user may first backload I-beam **964** including a wire coupled to a proximal portion of I-beam **964**, anvil **904**, and distal bodies **906, 907** through the distal end of a working channel of an endoscope, and move the wire and proximal portions of anvil **904** and distal bodies **906, 907** out of a proximal end of the endoscope prior to inserting apparatus **901** into a patient's body. Then, in some examples, the user may grasp tissue using a grasper or other tool to move tissue between anvil **904** and distal bodies **906, 907**. Once tissue is positioned in the active region of apparatus **901**, a user may translate a wire coupled to I-beam **964** distally and cause I-beam **964** to push anvil **904** and distal bodies **906, 907** radially-inward due to the interface between top portion **970** and bottom portion **971** on the one hand and anvil **904** and distal bodies **906, 907** on the other hand. Moving I-beam **964** distally may cause anvil **904** and distal bodies **906, 907** to close or clamp down onto tissue. As I-beam **964** is pushed distally towards tissue, edge **966** may engage and cut tissue. When cartridge **910** is positioned between top portion **970** and bottom portion **971**, actuation ramps **968, 969** may move into a cavity of cartridge **910** and contact one or more pistons, similar to pistons or spaces **751** shown in FIG. **7**. As a user moves I-beam **964** distally, actuation ramps **968, 969** may deploy fasteners from openings **914** by moving pistons within cartridge **910**. As the user continues to move I-beam distally, edge **966** and intermediate portion **965** may be positioned within channel **916**. Since anvil **904** and distal bodies **906, 907** are biased radially outward, anvil and distal bodies **906, 907** may be slidably engaged with I-beam **964** to hold anvil **904** and distal bodies **906, 907** in a closed, clamped position while fasteners are deployed from cartridge **910**.

(38) Each of the aforementioned apparatuses and devices may be used to grasp, couple, and/or cut tissue. In some examples, a user may load a proximal portion of a stapler device in a working channel of an endoscope by backfeeding the proximal portion through a distal end of an endoscope working channel to position a portion of the elongate body within the working channel. Once the proximal portion is positioned within a working channel, a handle assembly with one or more actuation devices may be coupled to the proximal end of the apparatus. The user may then introduce the endoscope into the patient's body and move the endoscope towards a target area. The

user may locate a target area (such as a tumor or other diseased tissue) present in a body lumen of a subject using the endoscope by directly visualizing the target area using an image sensor. Once the user has positioned the endoscope's distal end proximate to a target area, the user may position a tissue acquisition tool within a working channel of the endoscope, if such a tool is not already present. When the distal end of the endoscope is positioned at the target area, the user may then position tissue between or close to the active portion of the apparatus, e.g. the space between the apparatus's anvil and distal bodies. Once tissue is positioned within the stapler device's active portion, the user may move the stapler device's anvil and distal bodies to a closed position and clamp down on the grasped tissue with the apparatus. The user may then actuate an actuator to deploy fasteners into the clamped tissue and against the stapler device's anvil. In some examples, the user may actuate a knife in the apparatus to cut portions of the target tissue either before, during, or after fastening tissue together via fasteners.

(39) By providing a stapler instrument that a user may replace an empty staple cartridge with a stapling cartridge full of fasteners without having to remove the stapler instrument from a patient's body during a procedure, a user may reduce overall procedure time and avoid unnecessary harm to a patient's body caused by the repeated removal and re-insertion of a stapler instrument when replacing an empty stapling cartridge.

(40) It will be apparent to those skilled in the art that various modifications and variations may be made in the disclosed devices and methods without departing from the scope of the disclosure. Other aspects of the disclosure will be apparent to those skilled in the art from consideration of the specification and practice of the features disclosed herein. It is intended that the specification and examples be considered as exemplary only.

## Claims

1. A fastening device comprising: a flexible longitudinal body including a central longitudinal axis and a distal end; an anvil extending distally from the distal end of longitudinal body; and a distal body extending distally from the distal end of the longitudinal body, the distal body configured to removably receive a cartridge having a plurality of fasteners, wherein the distal body includes a first distal body and a second distal body separated by a gap configured to receive the cartridge; wherein at least one of the anvil and the distal body is flexible and has a first configuration with a distal portion of the at least one of the anvil and the distal body extending radially outward from the central longitudinal axis and a second configuration with the distal portion closer to the central longitudinal axis than in the first configuration.
2. The device of claim 1, wherein the fastening device includes a protrusion extending from a surface of anvil, wherein the protrusion is configured to engage the distal body.
3. The device of claim 1, wherein the at least one of the anvil and the distal body is the anvil; and wherein the distal body is flexible and has a first configuration with a distal portion of the distal body extending radially outward from the central longitudinal axis and a second configuration with the distal portion of the distal body closer to the central longitudinal axis than in the first configuration.
4. The device of claim 3, wherein each of the distal body and the anvil is curved in the respective first configuration and substantially straight in the second configuration.
5. The device of claim 4, wherein the cartridge includes an actuation wire extending proximally from a proximal end of the cartridge through a channel of the longitudinal body; and wherein the actuation wire is configured to move proximally relative to the cartridge and the anvil to deploy fasteners from the cartridge.
6. The device of claim 4, wherein the distal body includes a first distal body and a second distal body spaced from the first distal body, wherein the cartridge includes a distal protrusion extending distally from a distal surface of the cartridge, and wherein the distal protrusion is configured to be

received between distal ends of the first distal body and the second distal body.

7. The device of claim 1, further comprising the cartridge removably coupled to the distal body.

8. The device of claim 7, wherein the distal body includes a groove configured to slidably receive a flange of the cartridge.

9. The device of claim 7, wherein a distal portion of the cartridge includes a locking portion configured to be received by a locking gap of the distal body.

10. The device of claim 1, further comprising a flexible catheter configured to translate relative to the longitudinal body, wherein the flexible catheter moves one or more of the anvil and the distal body towards each other when the flexible catheter is moved distally relative to the longitudinal body.

11. The device of claim 1, further comprising: an I-beam including a top portion, a bottom portion, a distal-facing sharp edge and at least one actuation ramp, and the cartridge, wherein the cartridge is configured to receive the at least one actuation ramp, wherein the I-beam moves proximally and distally relative to the anvil and the distal body.

12. The device of claim 11, wherein the distal-facing sharp edge is proximal to a distal end of the I-beam and extends between the top portion and the bottom portion.

13. The device of claim 1, further comprising a cartridge configured to hold a plurality of fasteners and including a proximal elongate extending proximally from a proximal end of the cartridge, wherein the proximal elongate includes a lumen and at least one actuation wire positioned within the lumen, and wherein the actuation wire is configured to deploy fasteners from the cartridge and/or actuate a knife within the cartridge.

14. The device of claim 1, wherein each of the distal body and the anvil includes a radially-inward facing surface having at least one of a roughened surface, serrations, and teeth.

15. A tissue fastening device comprising: a flexible tube; a fastener device positioned within a lumen of the flexible tube, the fastener device comprising: a flexible anvil that has a first configuration with a distal portion of the anvil extending radially outward from a central longitudinal axis of the fastener device and a second configuration with the distal portion of the anvil closer to the central longitudinal axis of the fastener device than in the first configuration, wherein the anvil includes a protrusion extending from a surface of the anvil; a flexible distal body configured to removably receive a cartridge of fasteners, wherein the flexible distal body has a first configuration with a distal portion of the flexible distal body extending radially outward from the central longitudinal axis of the fastener device and a second configuration with the distal portion closer to the central longitudinal axis of the fastener device than in the first configuration, wherein the protrusion is configured to engage the flexible distal body; wherein the tube is configured to translate distally relative to the fastener device to transition each of the anvil and the flexible distal body from the respective first configuration to the respective second configuration.

16. The device of claim 15, further comprising the cartridge removably coupled to the flexible distal body.

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